

Laboration 1 Kit

- We have provided you with a Laboration Kit (a ZIP file)
 - That includes a single python file
 - DT075A Lab1.py
 - Which performs regression on a simple mathematical function
- In detail the code has code for:
 - Creating a simple Machine Learning model
 - A minimal training dataset
 - Code for starting the training
 - A dataset for testing the model
 - Code for testing the model
 - Code for plotting the results

Libraries and imports

```
import tensorflow as tf  
import numpy as np  
import matplotlib.pyplot as plt
```

- Tensorflow for all ML functions (including keras)
- Matplotlib for plotting graphs
- Numpy for NumPy Arrays

Model

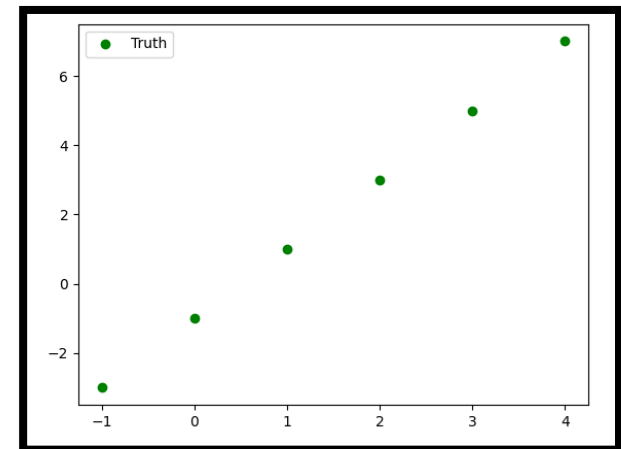
```
#Model
model = tf.keras.models.Sequential([
    tf.keras.layers.Dense(units=2, input_dim=1, activation='linear'),
    tf.keras.layers.Dense(units=1, activation='linear')
])
```

- A ML model using Keras sequential API
- An input layer, taking 1 input (x), 2 neurons wide
- An output layer producing 1 output (y), 1 neuron wide
- Linear activation functions

The Training Dataset

```
#Formula:  $Y = 2X - 1$ 
train_input_X = np.array([-1.0, 0.0, 1.0, 2.0, 3.0, 4.0])
train_output_Y = np.array([-3.0, -1.0, 1.0, 3.0, 5.0, 7.0])
```

- 5 Touples
 - Which includes paired x and y values
 - (X,Y) : (-1,-3) (0,-1) (1,1) (2,3) (3,5) (4,7)
- Created from the function $Y = 2X - 1$



Training the model

```
#Train the model  
~~~~~  
model.fit(train_input_X, train_output_Y, epochs=5)
```

- Train using the training dataset (x and y values)
- Train for 5 epochs

Testing the model

```
#Test the model  
test_input_X = np.array([-3.0, -2.0, -1.0, 0.0, 1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0])  
test_prediction_Y = model.predict(test_input_X)
```

- Create a testing dataset
 - X from -3 to 10
- Use our trained model to predict these X values

Plot the results

```
#Plot the results
plt.scatter(test_input_X, test_prediction_Y, color='red', label='Prediction')
plt.plot(train_input_X, train_output_Y, color='green', label='Truth')
plt.legend()
plt.show()
```

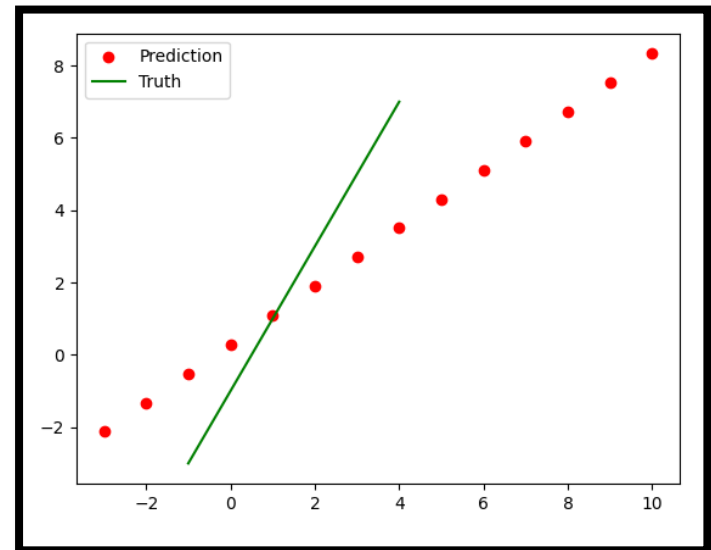
- Scatterplot red dots for each prediction (the results)
- Line plot a green line for the ground truth (the training data)

Running The Code For the First Time

- If everything has been installed correctly
 - The code should work out of the box
 - Giving you a quite poor regression system (not the same every time)

```
Epoch 1/5  
1/1 [=====] - 0s 381ms/step - loss: 24.6514  
Epoch 2/5  
1/1 [=====] - 0s 996us/step - loss: 16.7974  
Epoch 3/5  
1/1 [=====] - 0s 962us/step - loss: 12.2355  
Epoch 4/5  
1/1 [=====] - 0s 2ms/step - loss: 8.9382  
Epoch 5/5  
1/1 [=====] - 0s 1ms/step - loss: 6.3789
```

- You might also get some warnings
 - Because of GPU acceleration etc.
 - Ignore these...



Improving the Code

- Observe the results and plots
 - Explore and understand what this means
 - Try to learn a more complex formulas
 - Some can be quite hard, since the datasets are quite small...
- Tips on what you could do:
 - Let the system train longer (increase epochs)
 - Make the model wider (increase layer size)
 - Make the model deeper (add more layers)
 - Change the optimizer (ex. adam, ftrl)
 - Change the loss function (ex. mae, huber)
 - Change the activation function (ex. relu, sigmoid, softmax)